

Winning Space Race with Data Science

P R E S E N T A T I O N

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OUTLINE

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Github: <https://github.com/MariuszGaw/IBM-Data-Science-Project>

EXECUTIVE SUMMARY

SUMMARY OF METHODOLOGIES

1. Data Collection through API
2. Data Collection with Web Scraping
3. Data Wrangling- Exploratory Data Analysis with SQL
4. Exploratory Data Analysis with Data Visualization
5. Interactive Visual Analytics with Folium
6. Machine Learning Prediction

SUMMARY OF ALL RESULTS

1. Exploratory Data Analysis Result
2. Interactive Analytics and Visualization
3. Predictive Analytics

INTRODUCTION

PROJECT BACKGROUND AND CONTEXT

The commercial space age is here, with companies making space travel more affordable. SpaceX, Virgin Galactic Rocket Lab and Blue Origin manufactures reusable rockets. SpaceX is the most successful, with achievements like sending spacecraft to the ISS, launching Starlink for satellite internet, and conducting crewed missions. One key reason for SpaceX's success is the relatively low cost of their rocket launches - \$62 million for a Falcon 9 launch compared to over \$165 million from other providers. A big part of the cost savings comes from reusing the rocket's first stage.

PROBLEMS I WANT TO FIND ANSWERS

- Identifying all key factors that affect the landing outcome.
- Analyzing how individual variables are related to each other and how they influence the final result.
- Determining the optimal conditions that maximize the probability of a successful landing.

SECTION 1

Methodology

Methodology

1

Data collection / Wrangling / Formatting

API

Web scraping

2

Exploratory data analysis

Python (Pandas, NumPy)

SQL

3

Data Visualization

Matplotlib / Seaborn / Folium / Dash

4

Machine learning

Logistic regression / Support vector machine (SVM)

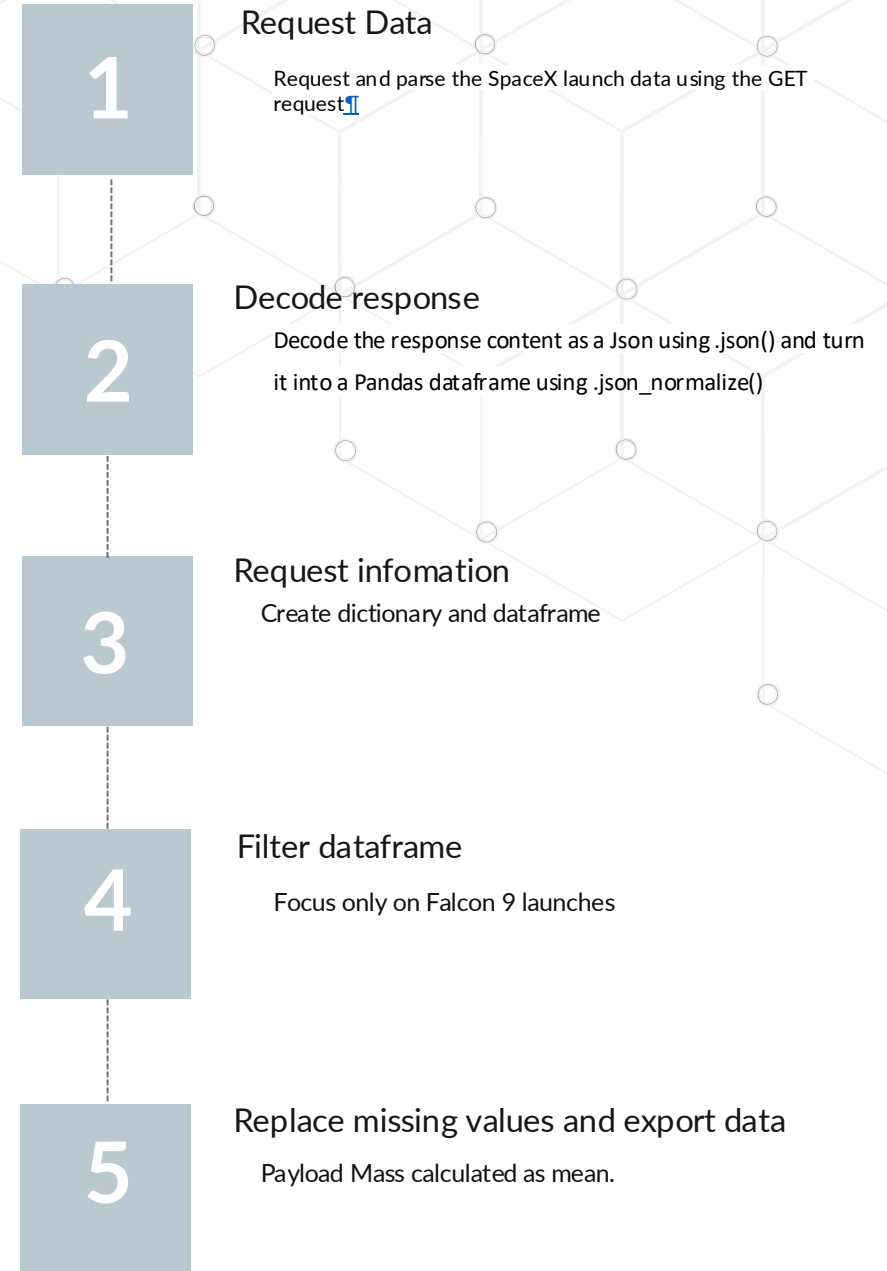
Decision tree / K nearest neighbors (KNN)

Data Collection

SpaceX - API

The data will come from the SpaceX REST API (api.spacexdata.com/v4/), specifically from the `/launches/past` endpoint, which provides information on past launches, including rocket details, payload, landing specifications, and outcomes.

We'll retrieve this data using a GET request with Python's requests library and parse the JSON response into a DataFrame using `json_normalize`. This will allow us to work with structured data in a table format for analysis.

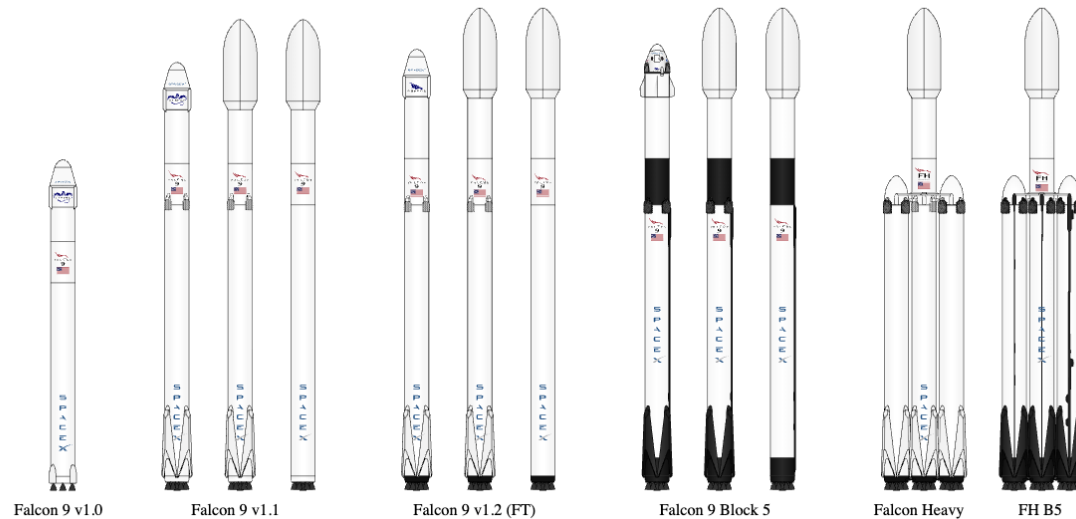


Data Collection Web Scrapping

Web scrapping to collect Falcon 9 historical launch records from a Wikipedia page titled List of Falcon 9 and Falcon Heavy launches

Access:

https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches



1

Request data

Request the Falcon9 Launch Wiki page from its URL

2

Creat BeautifulSoup Object

Create a `BeautifulSoup` object from the HTML `response`

3

Extract data

Extract all column/variable names from the HTML table header

4

Create data frame

Create a data frame by parsing the launch HTML tables

5

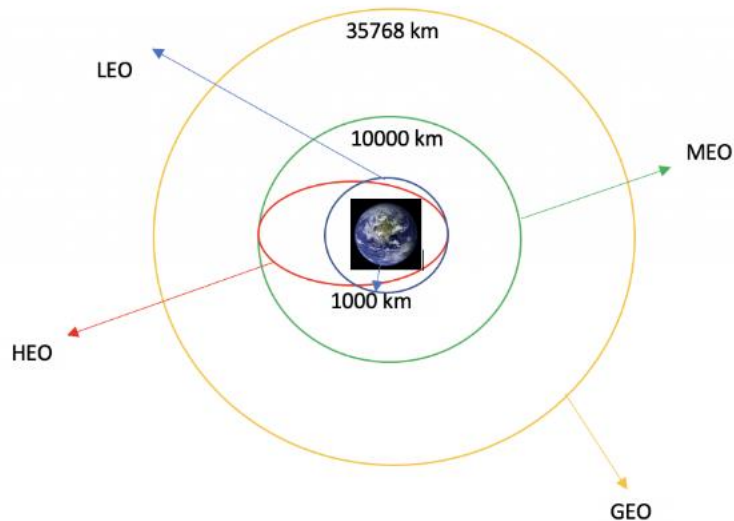
Create a CSV file

Export data do csv file

Data Wrangling

On this stage I will use Exploratory Data Analysis (EDA) to find some patterns in the data and determine what would be the label for training supervised models.

Import Libraries and Define Auxiliary Functions [1](#) (Pandas, Numpy)



1

Import

Import Libraries and Define Auxiliary Functions [1](#)

2

Number of launches

Calculate the number of launches on each site

3

The number and occurrence of each orbit

Determine the number and occurrence of each orbit in the column

4

The number and occurrence of mission outcome of the orbits

Use the method `.value_counts()` on the column Outcome to determine the number of landing_outcomes

5

Landing outcome label from Outcome column

Using the Outcome, create a list where the element is zero if the corresponding row in Outcome is in the set `bad_outcome`; otherwise, it's one

Exploratory Data Analysis (EDA) with SQL

DISPLAY	the names of the unique launch sites in the space mission
DISPLAY	5 records where launch sites begin with the string 'CCA'¶¶
DISPLAY	the total payload mass carried by boosters launched by NASA (CRS)
DISPLAY	average payload mass carried by booster version F9 v1.1
LIST	the date when the first successful landing outcome in ground pad was achieved¶¶
LIST	the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000¶¶
LIST	the total number of successful and failure mission outcomes
LIST	the names of the booster versions which have carried the maximum payload mass
LIST	the records which will display the month names, failure landing outcomes in drone ship ,booster versions, launch site for the months in year 2015.¶¶
RANK	the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

Exploratory Data Analysis (EDA) with Data Visualization

Scatter point chart

Visualize the relationship between Flight Number and Launch Site

Scatter point chart

Visualize the relationship between Payload Mass and Launch Site

Bar chart

Visualize the relationship between success rate of each orbit type

Scatter point chart

Visualize the relationship between FlightNumber and Orbit type

Scatter point chart

Visualize the relationship between Payload Mass and Orbit type

Line chart

Visualize the launch success yearly trend

Interactive Map with Folium

To present the launch data on an interactive map, latitude and longitude coordinates were used for each launch site. Circle markers were placed on the map and labeled with the corresponding launch site names.

Launch outcomes were categorized as success or failure and encoded as classes 1 and 0, represented by green and red markers within a MarkerCluster. Additionally, the formula was applied to calculate distances between launch sites and nearby landmarks. This analysis was used to determine how close the launch sites are to:

- railways,
- highways,
- coastlines,
- and nearby cities.



Dashboard with Plotly

The primary objective of the analysis was to identify which feature values are associated with the highest number of successful launches. To support this goal, an interactive dashboard was developed using Plotly Dash, allowing users to explore the data dynamically.

The dashboard includes pie charts illustrating the percentage of successful and failed launches across different launch sites, highlighting the usefulness of the launch site as a predictive feature.

Additionally, scatter plots were created to visualize the relationship between payload mass and landing outcome, with color encoding based on the booster version used. This visualization also helps identify the payload mass ranges most commonly associated with successful missions.



Machine Learning Prediction with Scikit-learn

Functions from the Scikit-learn library were used to build the machine learning models. The prediction process consisted of several key steps, including data standardization and splitting the dataset into training and testing subsets.

Multiple machine learning algorithms were implemented, namely:

- logistic regression,
- support vector machines (SVM),
- decision trees,
- and k-nearest neighbors (KNN).

Each model was trained on the training data, followed by hyperparameter tuning to identify the optimal configurations. Finally, the models were evaluated using accuracy metrics and confusion matrices to assess their performance.

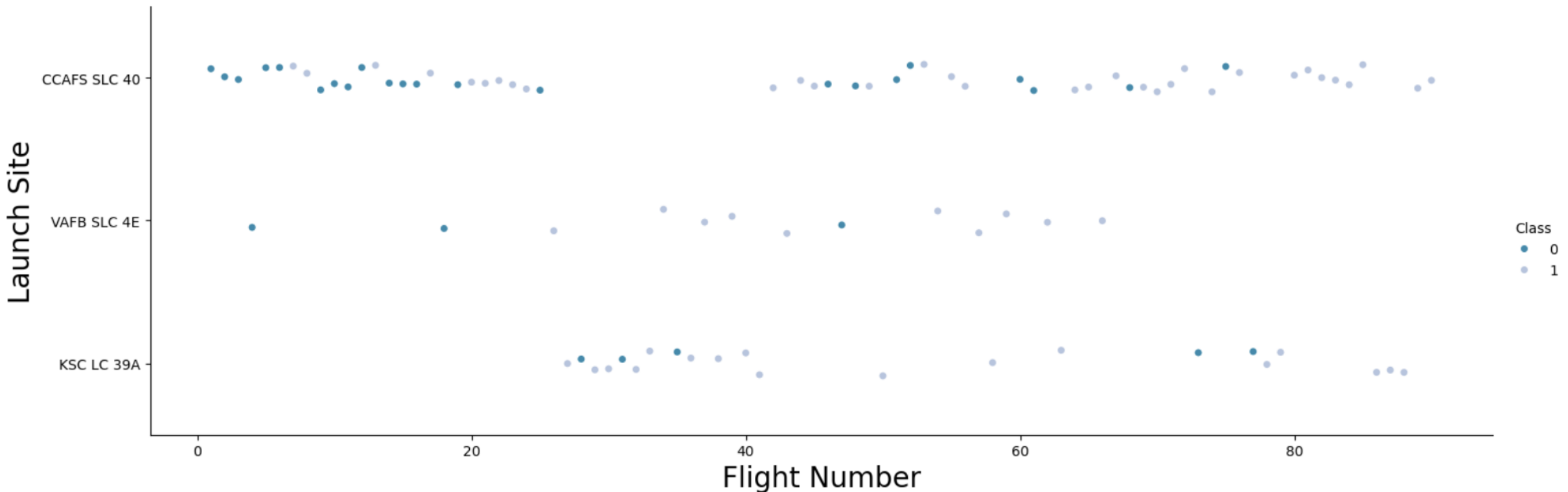


SECTION 2

Insights drawn from Exploratory Data Analysis (Data Visualization and SQL)

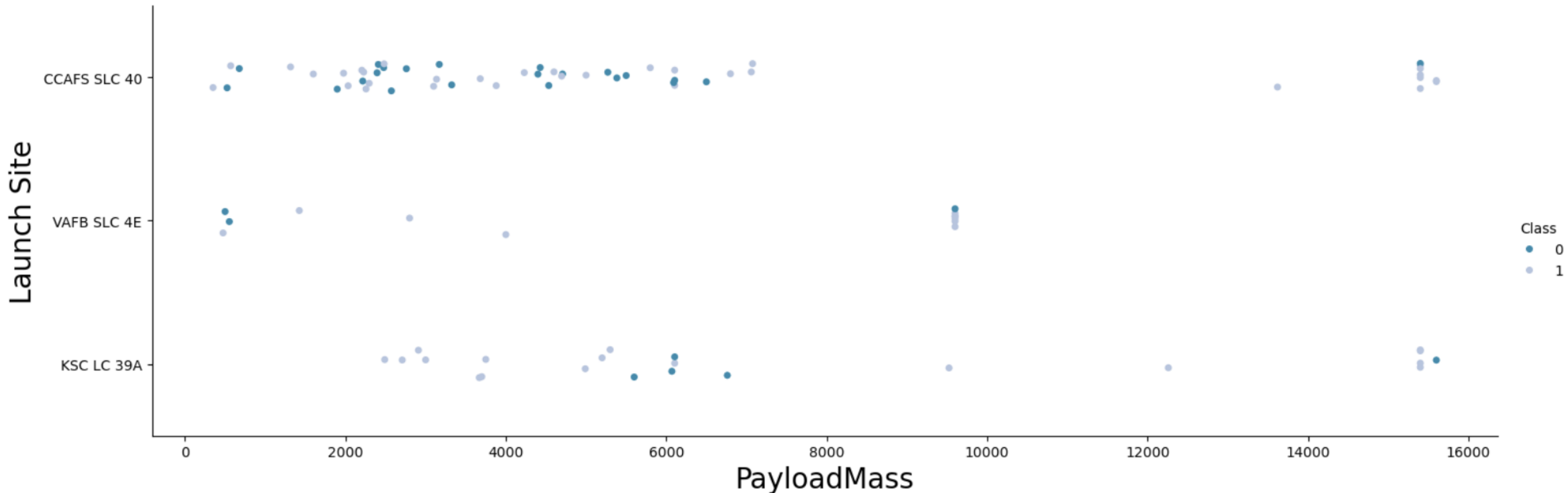
Flight Number vs. Launch Site

From my observation I noticed that the probability of success rate growth with the number of flight. The first launch site, which is the oldest in the given time (CCAFS SLC 40) had the worst ratio between success and failed launch. In comparison the others much often ended positive.



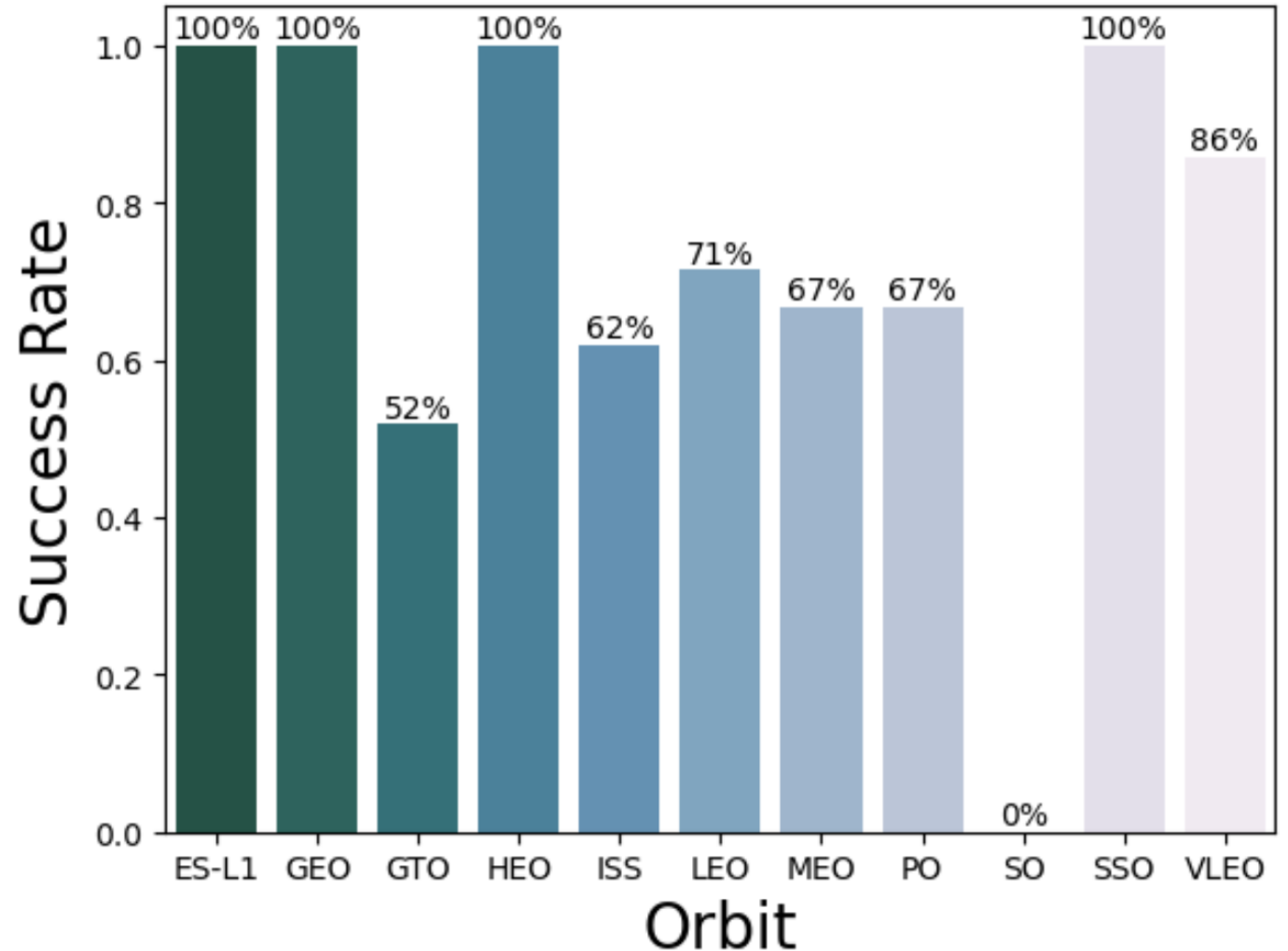
Payload vs. Launch Site

The given scatter plot shows relation between Launch Site and Payload Mass. As a mean feature we can observe that the bigger Payload, the highest chance on positive launch. Super heavy launch site has almost 100% success ratio.



Success Rate vs. Orbit Type

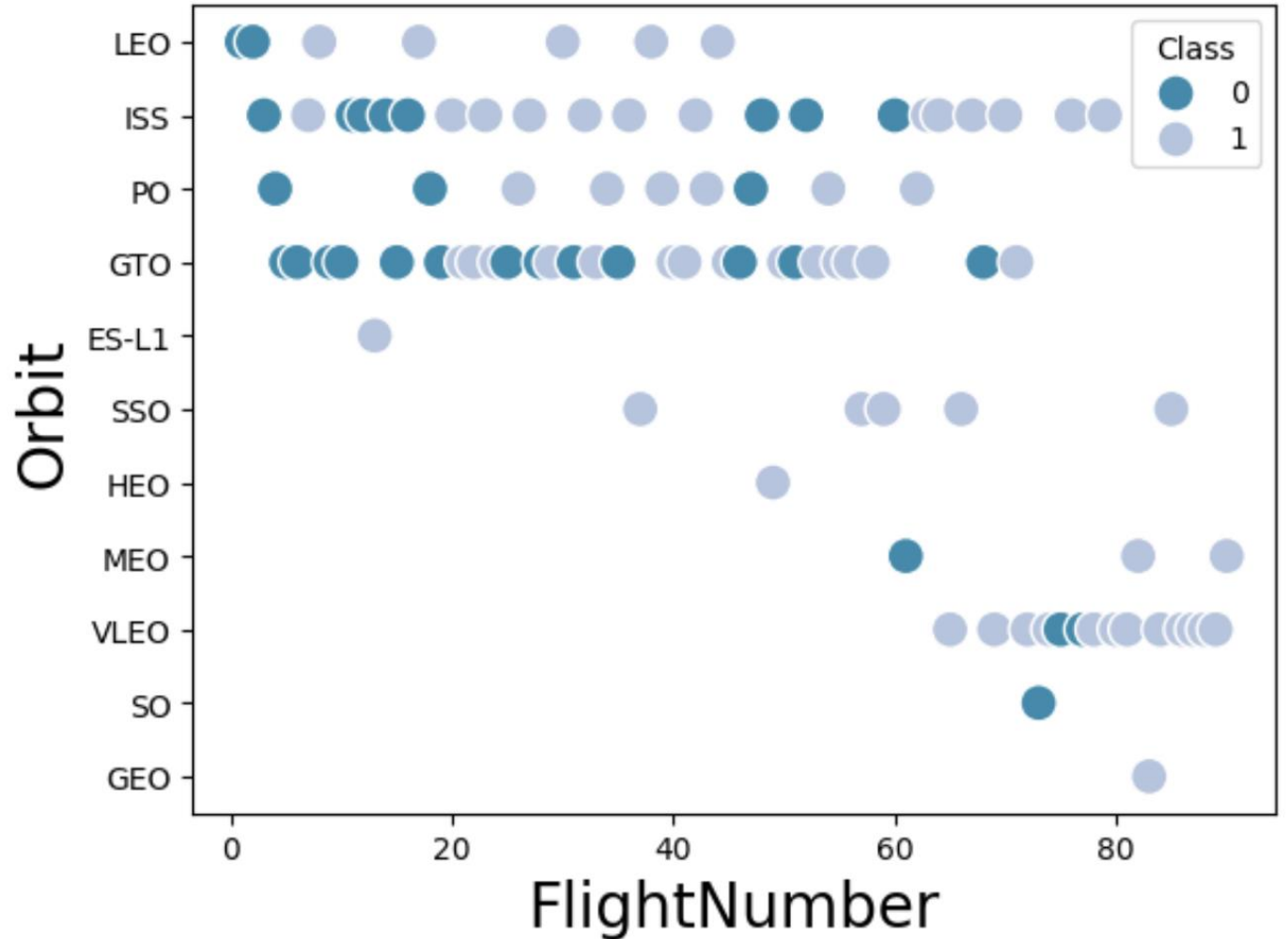
- There are 4 orbits with 100% success rate: ES-L1, GEO, HEO, SSO
- Only SO orbit has noticed 0% success rate
- The remaining orbits, such as: GTO, ISS, LEO, MEO, PO has success rate above 50%



Flight Number vs. Orbit Type

In the LEO orbit, success seems to be related to the number of flights.

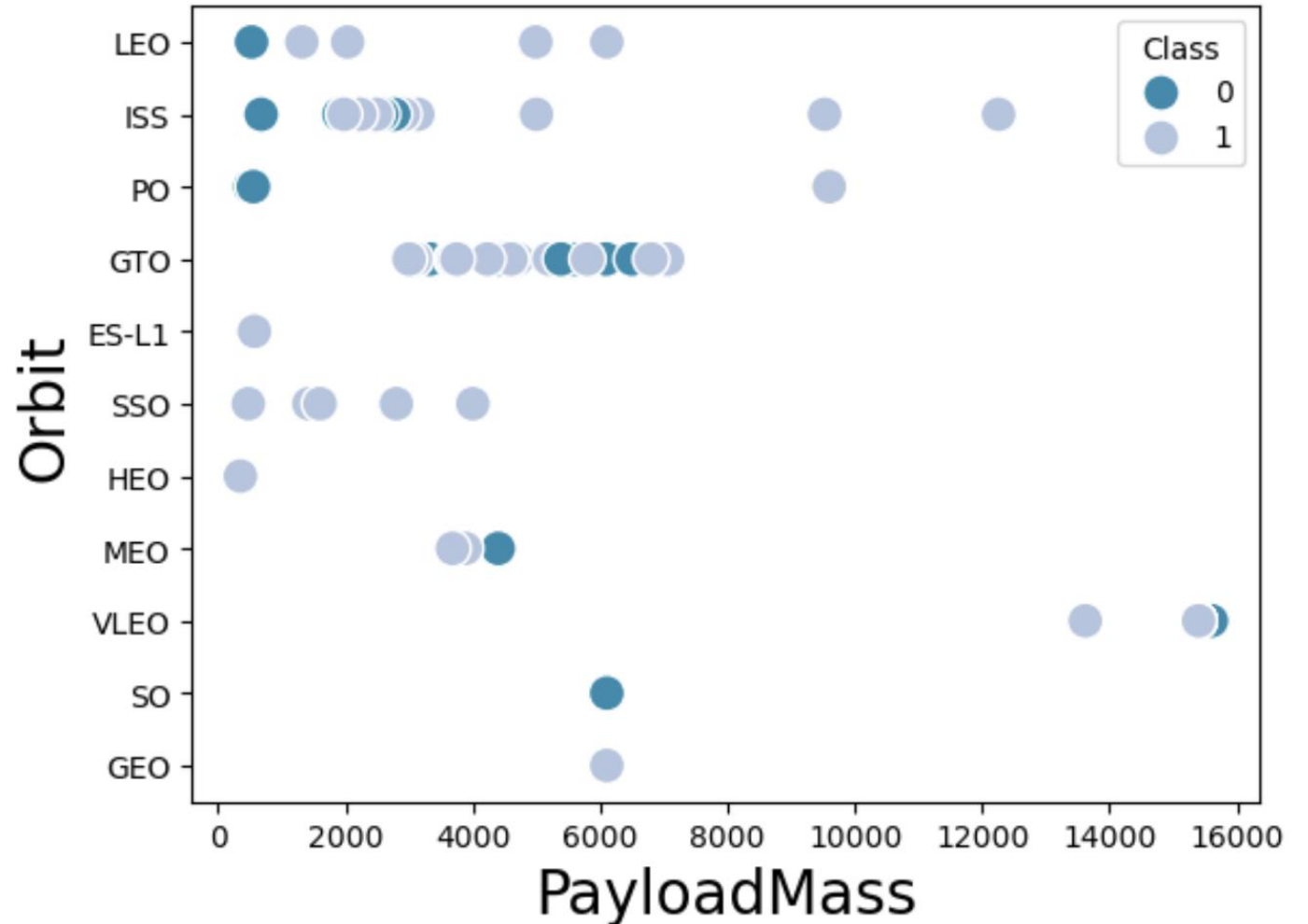
Conversely, in the GTO orbit, there appears to be no relationship between flight number and success.



Payload vs. Orbit Type

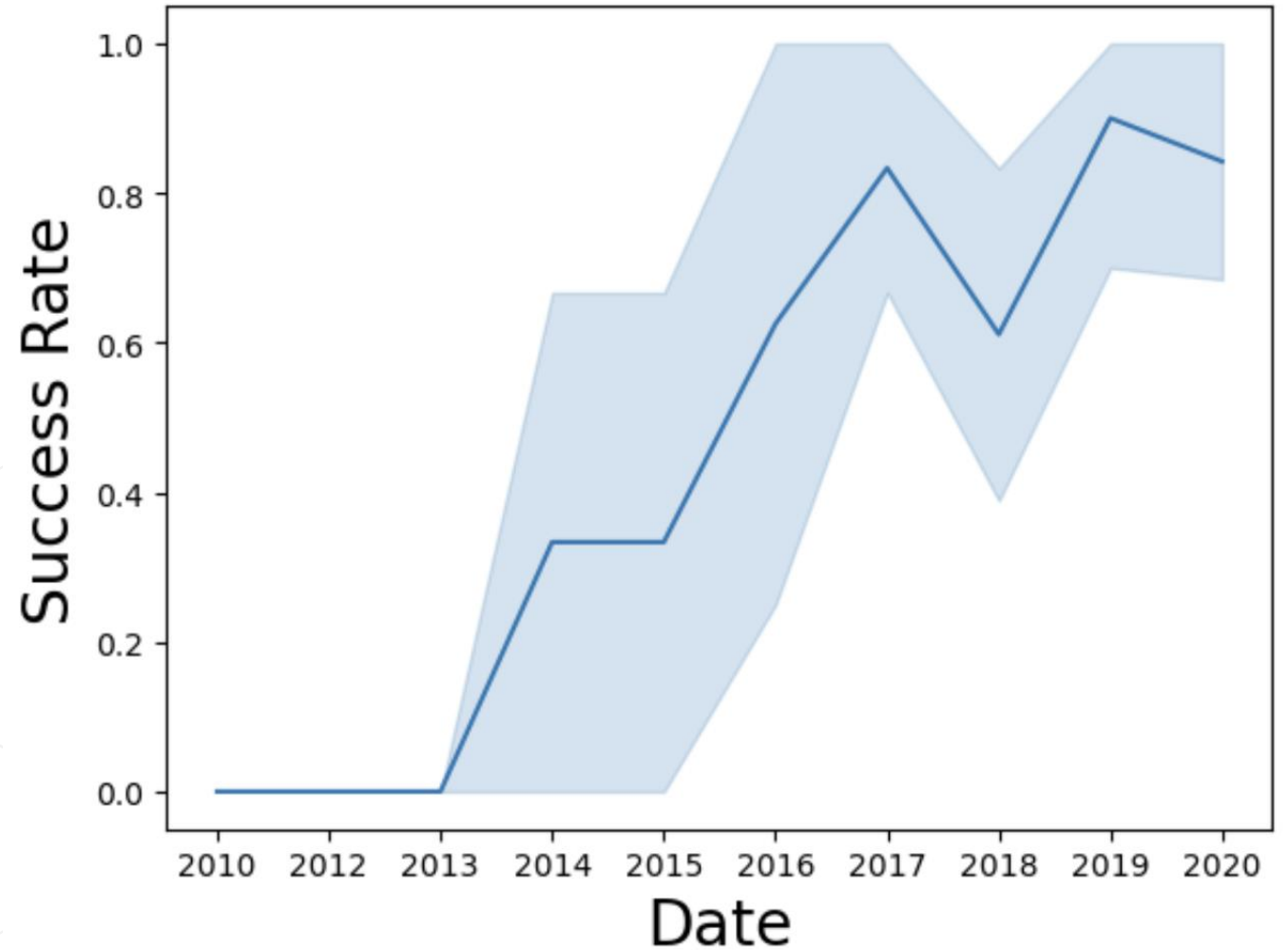
With heavy payloads the successful landing or positive landing rate are more for Polar, LEO and ISS.

However, for GTO, it's difficult to distinguish between successful and unsuccessful landings as both outcomes are present.



Launch Success Yearly Trend

We can observe that the success rate since 2013 kept increasing till 2020



All Launch Site Names

```
SELECT DISTINCT LAUNCH_SITE  
FROM SPACEXTBL
```

Launch_Site

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

Launch Site Names Begin with 'CCA'

```
SELECT *  
FROM SPACEXTBL  
WHERE LAUNCH_SITE LIKE 'CCA%' LIMIT 5
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

```
SELECT SUM(PAYLOAD_MASS__KG_)  
FROM SPACEXTBL  
WHERE CUSTOMER = 'NASA (CRS)'
```

SUM(PAYLOAD_MASS__KG_)

45596

Average Payload Mass by F9 v 1.1

```
SELECT AVG(PAYLOAD_MASS_KG_)  
FROM SPACEXTBL  
WHERE BOOSTER_VERSION = 'F9 v1.1'
```

AVG(PAYLOAD_MASS_KG_)

2928.4

First Successful Ground Landing Date

```
SELECT MIN(DATE)
FROM SPACEXTBL
WHERE LANDING_OUTCOME = 'Success (ground pad)'
```

MIN(DATE)

2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

```
SELECT PAYLOAD  
FROM SPACEXTBL  
WHERE LANDING_OUTCOME = 'Success (drone ship)' AND PAYLOAD_MASS_KG_ BETWEEN 4000 AND  
6000
```

Payload
JCSAT-14
JCSAT-16
SES-10
SES-11 / EchoStar 105

Total Number of Successful and Failure Mission Outcomes

```
SELECT MISSION_OUTCOME, COUNT(*) AS total_number  
FROM SPACEXTBL  
GROUP BY MISSION_OUTCOME
```

Mission_Outcome	total_number
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

```
SELECT BOOSTER_VERSION  
FROM SPACEXTBL  
WHERE PAYLOAD_MASS__KG_ =  
      (SELECT MAX(PAYLOAD_MASS__KG_) FROM  
      SPACEXTBL
```

Booster_Version

F9 B5 B1048.4

F9 B5 B1049.4

F9 B5 B1051.3

F9 B5 B1056.4

F9 B5 B1048.5

F9 B5 B1051.4

F9 B5 B1049.5

F9 B5 B1060.2

F9 B5 B1058.3

F9 B5 B1051.6

F9 B5 B1060.3

F9 B5 B1049.7

2015 Launch Records

```
SELECT SUBSTR(Date,6,2) AS month,Date,BOOSTER_VERSION,LAUNCH_SITE,[Landing_Outcome]
FROM SPACEXTBL
WHERE [Landing_Outcome] = 'Failure (drone ship)' AND SUBSTR(Date,0,5) = '2015'
```

month	Date	Booster_Version	Launch_Site	Landing_Outcome
01	2015-01-10	F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)
04	2015-04-14	F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
SELECT [Landing_Outcome], count(*) AS count_outcomes
FROM SPACEXTBL
WHERE DATE BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY [Landing_Outcome]
ORDER BY count_outcomes DESC
```

Landing_Outcome	count_outcomes
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

SECTION 3

Launch Sites Proximities Analysis

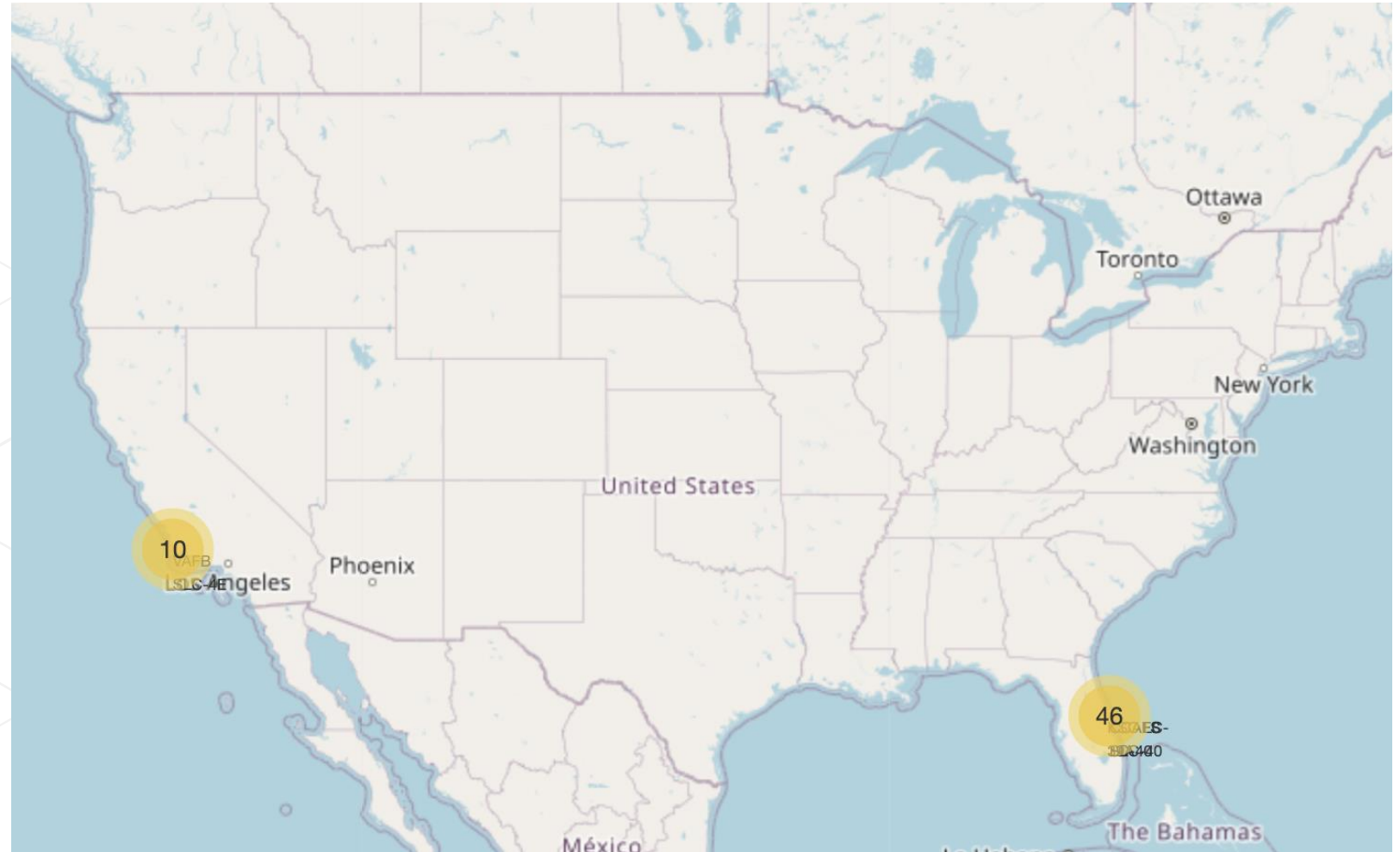
All launch sites on a map

The launch success rate may depend on many factors such as payload mass, orbit type, and so on. It may also depend on the location and proximities of a launch site, i.e., the initial position of rocket trajectories. Finding an optimal location for building a launch site certainly involves many factors and hopefully we could discover some of the factors by analyzing the existing launch site locations.

All launch sites on a map

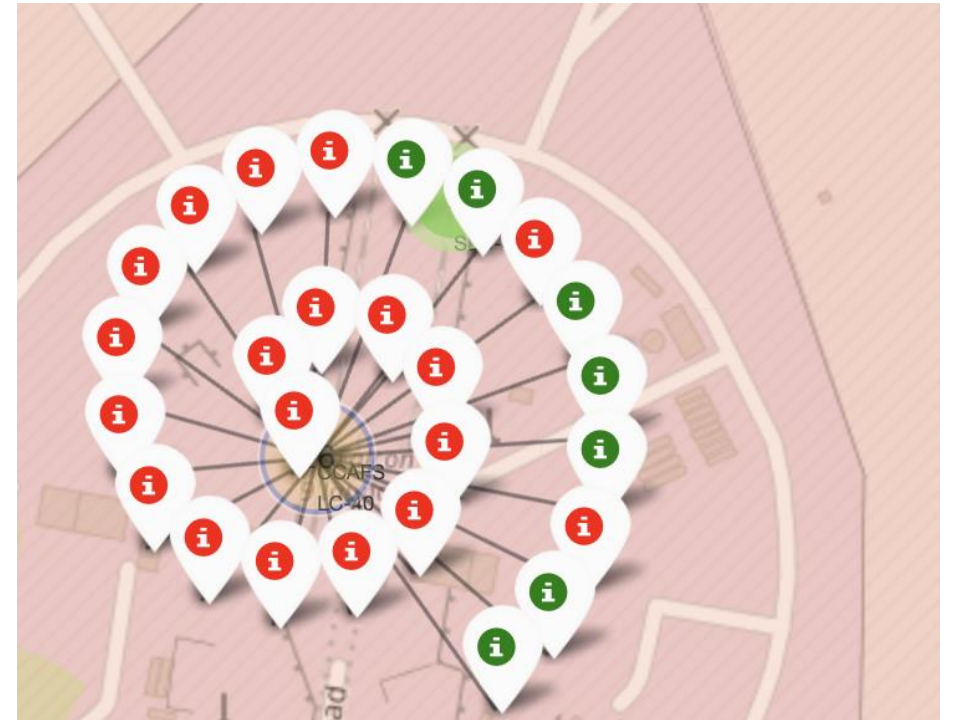
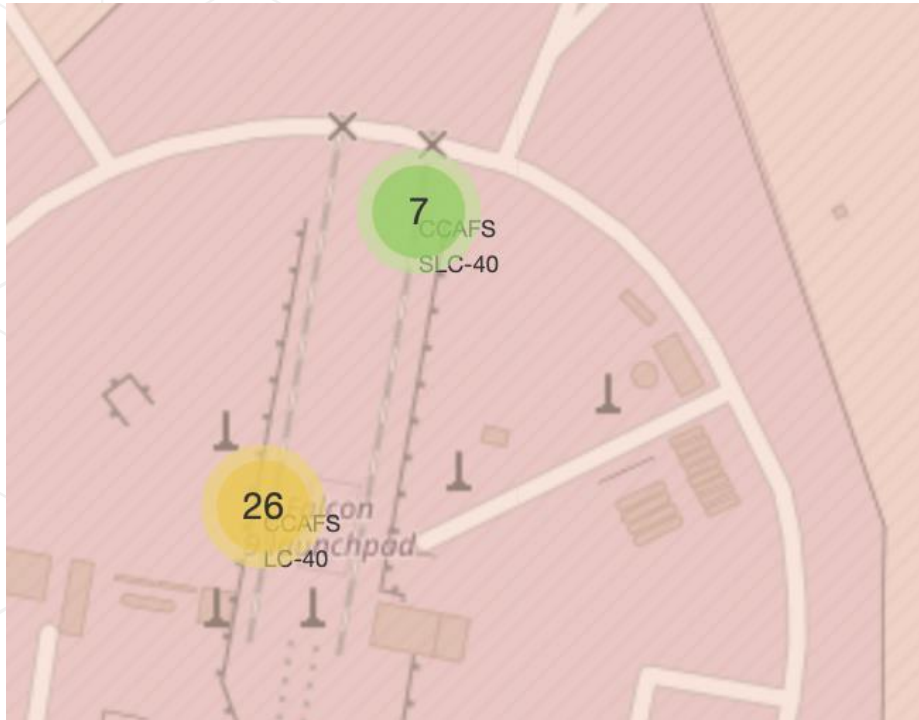
Most launch sites are near the Equator to take advantage of Earth's rotational speed, which helps with fuel savings and easier launches into equatorial orbits. It's worth to notice that many launch sites are near the coast for safety reasons,

	Launch Site	Lat	Long
0	CCAFS LC-40	28.562302	-80.577356
1	CCAFS SLC-40	28.563197	-80.576820
2	KSC LC-39A	28.573255	-80.646895
3	VAFB SLC-4E	34.632834	-120.610745



The success/failed launches for each site

From the color-labeled markers in marker clusters, we can easily identify which launch sites have relatively high success rates. For the launch site numbered CCAFS LC-40, marked on the map below, we see a positive indicator at a level of only 23%.

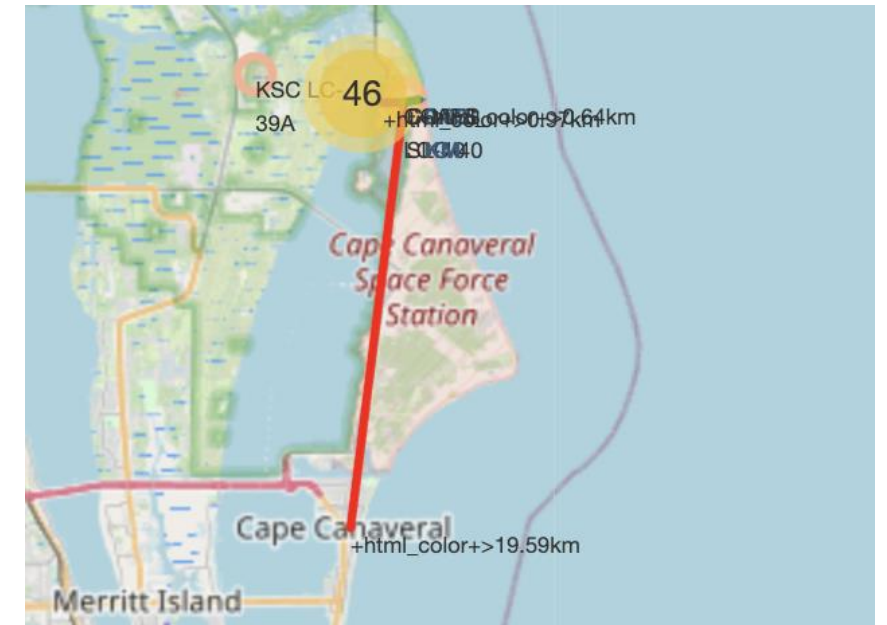


The distances between a launch site to its proximities

The distance from the surrounding area shows that proximity to the coast is crucial for determining the rocket launch site. In the case of a failed launch, there is a higher probability that the rocket will fall into the sea. Similarly, as we can see, the nearest city is significantly distant from this location, which further enhances safety and minimizes the risk to residents. The distance to the nearest infrastructure (highway and railway) from the launch site is small, which results from the need to ensure the transportation of rocket components.

Distance to nearest:

- highway – 0.64 km
- coastline – 0.93 km
- railway – 0.97 km
- city – 19.59 km



SECTION 4

Build a Dashboard with Plotly

Dash

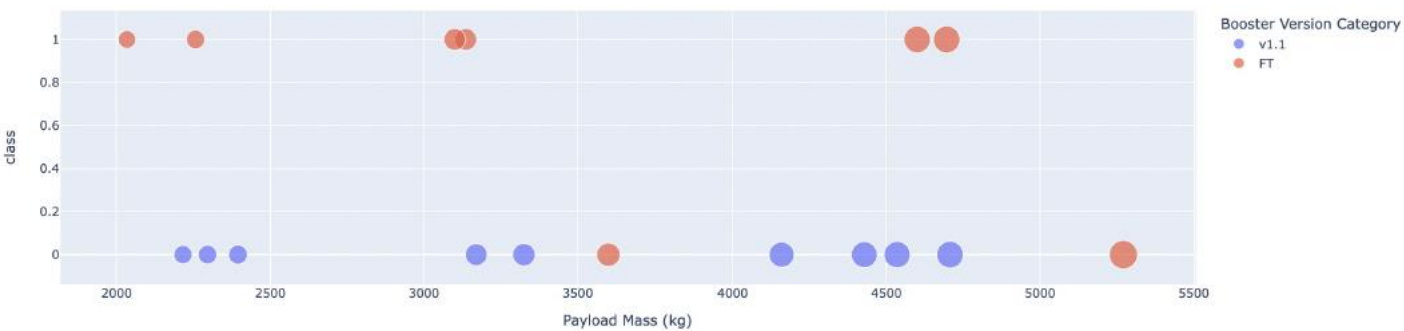
SpaceX Launch Records Dashboard

CCAFS LC-40

Total Success Launches for Site → CCAFS LC-40



Correlation Between Payload and Success for Site → CCAFS LC-40

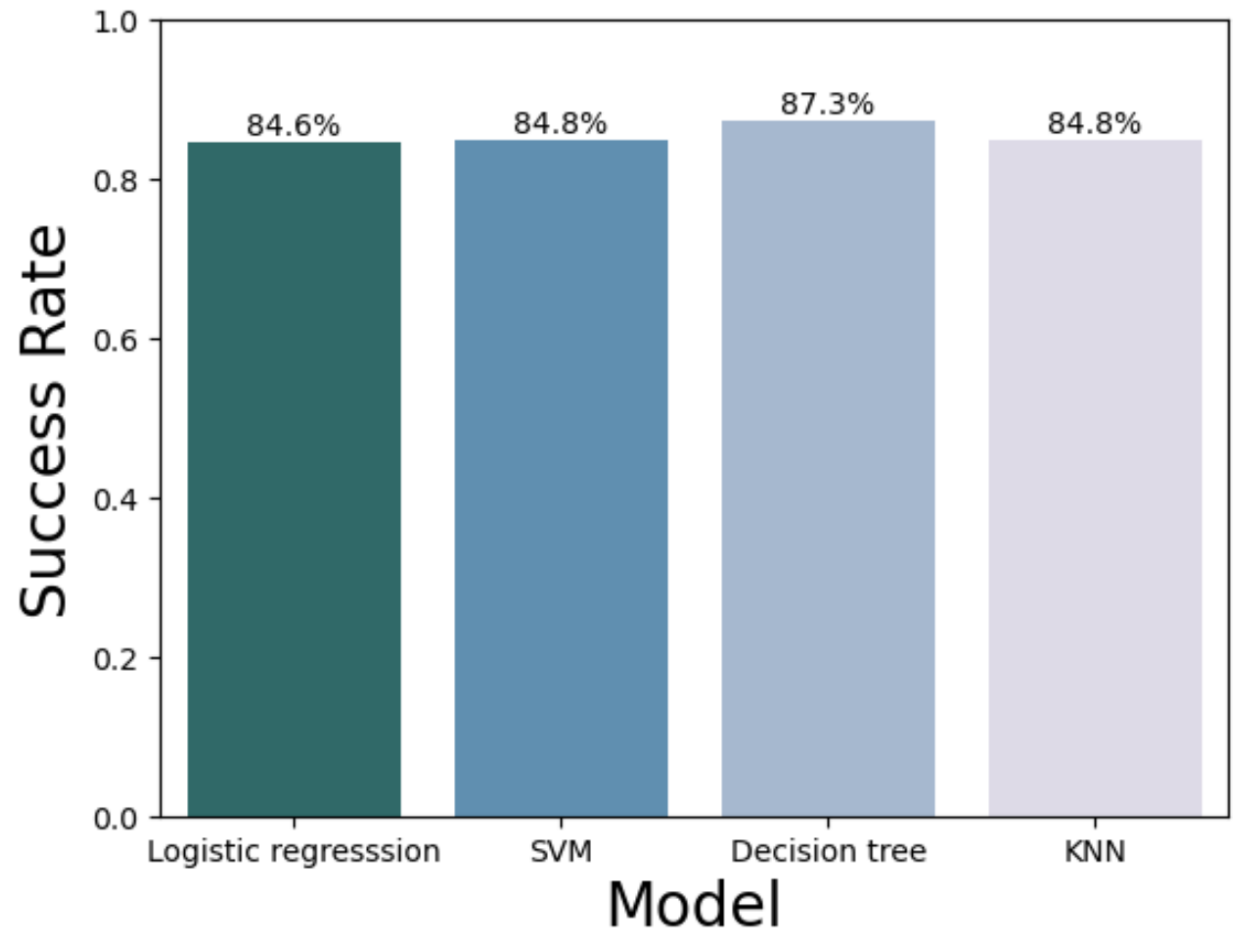


SECTION 5

Predictive Analysis (Classification)

The Machine Learning Prediction

Accuracy of the listed models in the field of machine learning.

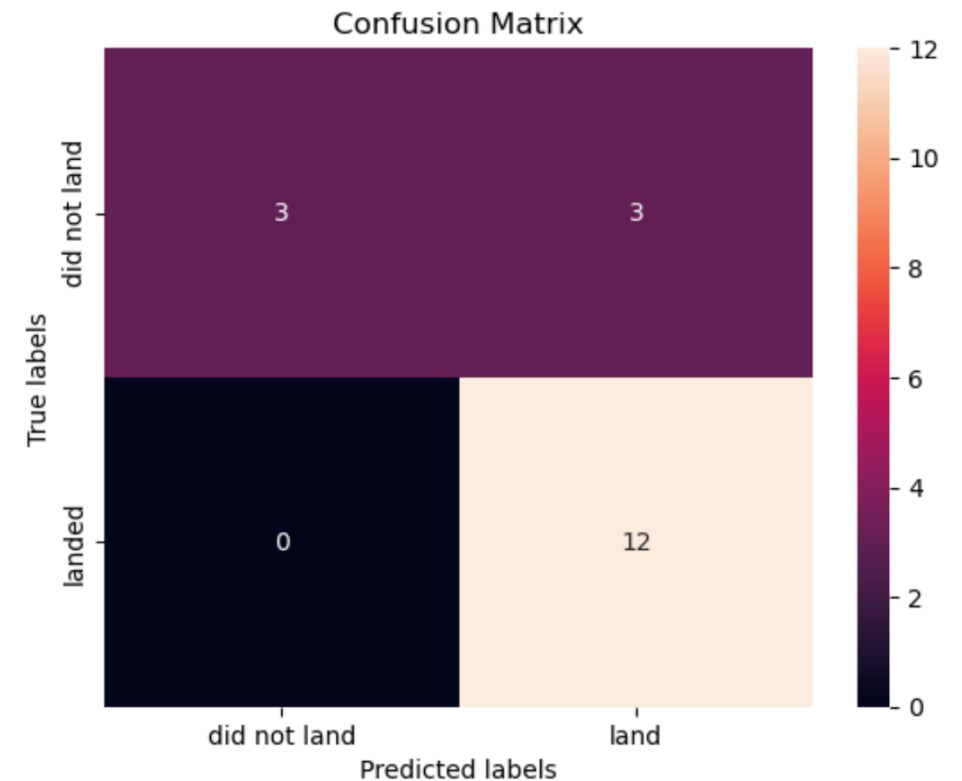


The Machine Learning Prediction

Logistic regression

Logistic regression is a supervised machine learning algorithm mainly used for classification problems, especially binary classification. It models the probability that an input belongs to a given class using the sigmoid (logistic) function, which outputs values between 0 and 1. Based on this probability, the model assigns a class label using a decision threshold (usually 0.5). Logistic regression is simple, fast, interpretable, and works well when the relationship between features and the log-odds of the outcome is approximately linear.

accuracy : 0.8464285714285713

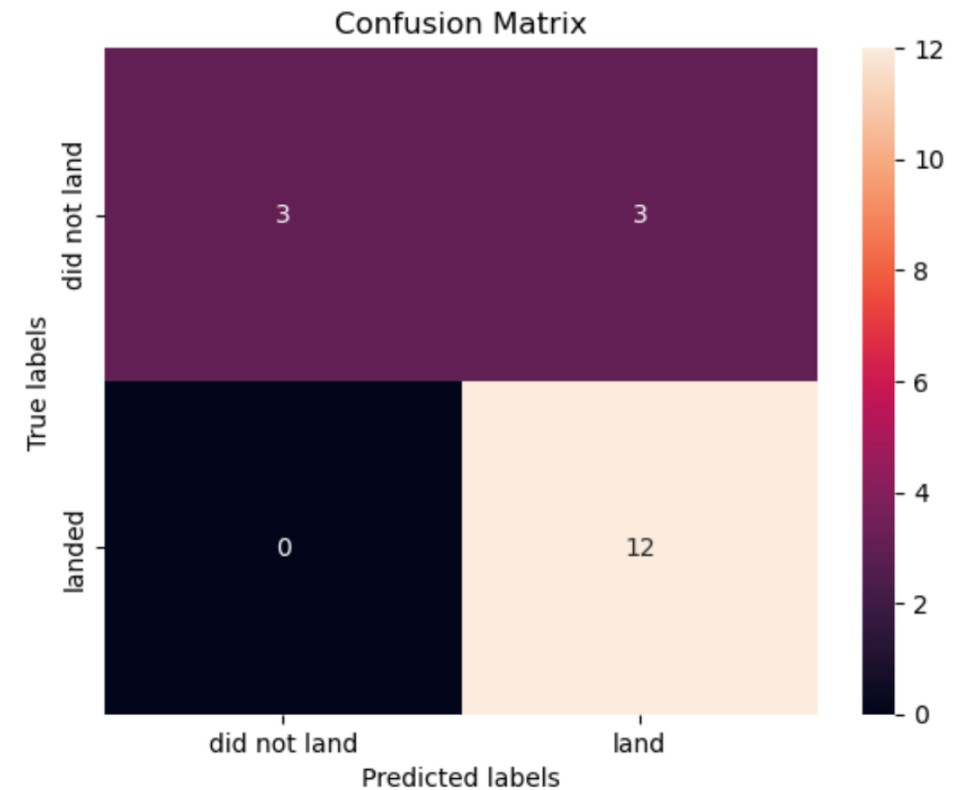


The Machine Learning Prediction

Support Vector Machine (SVM)

Support Vector Machine (SVM) is a supervised machine learning algorithm used for classification and regression tasks. Its main idea is to find an optimal decision boundary (hyperplane) that maximizes the margin between different classes. SVM focuses on the most important data points, called support vectors, which define this boundary. Thanks to kernel functions, SVM can model complex, non-linear relationships and often achieves high accuracy, especially in high-dimensional spaces.

accuracy : 0.8482142857142856

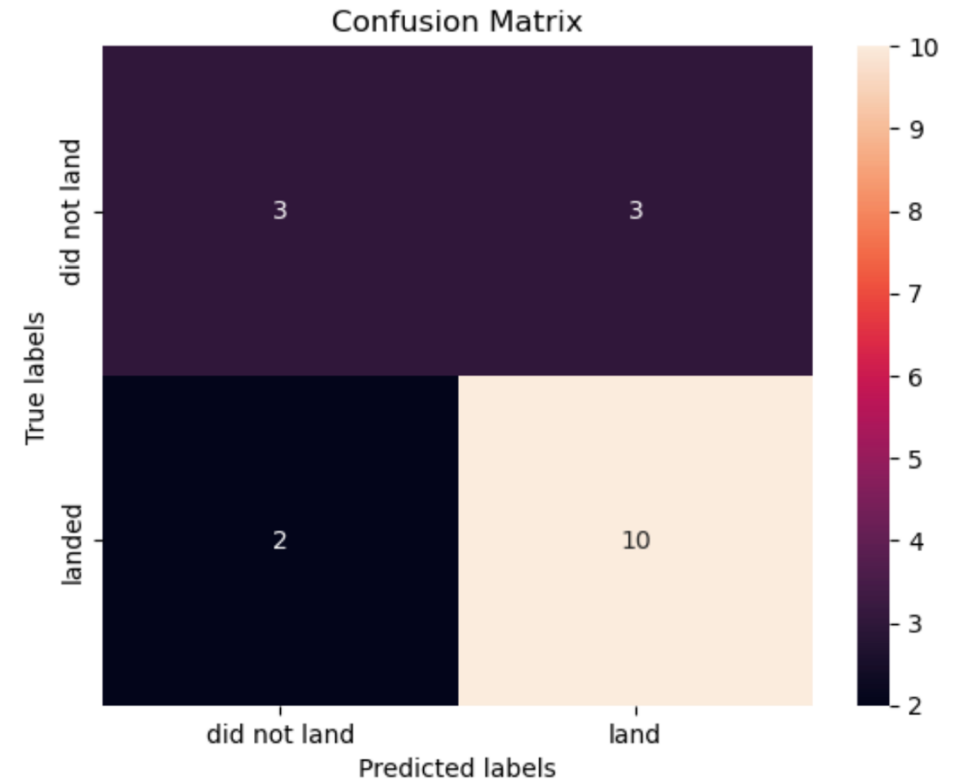


The Machine Learning Prediction

Decision Tree

Decision Tree is a supervised learning algorithm used for classification and regression. It works like a decision-making flowchart: at each step, the model splits the data based on a condition (e.g., "is feature $X > 5$?"), creating branches. Each node chooses the feature that best separates the classes using criteria like gini or entropy.

accuracy : 0.8732142857142857

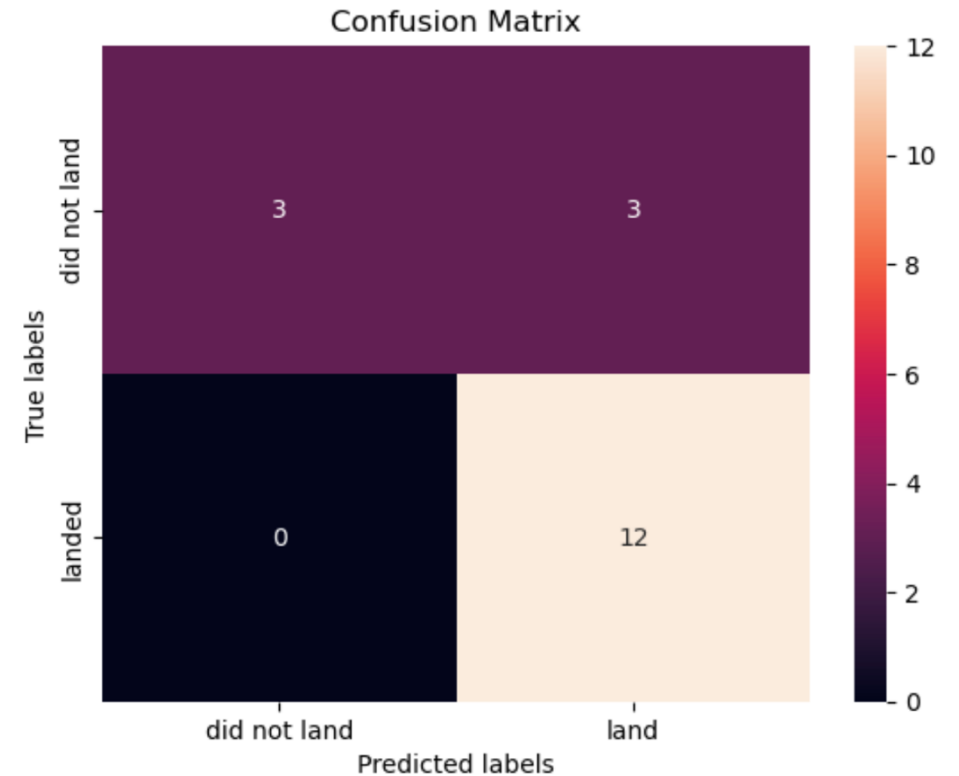


The Machine Learning Prediction

K nearest neighbors

K-Nearest Neighbors (KNN) is a supervised learning algorithm mainly used for classification, but also for regression. It works by comparing a new data point with the training data: for a new sample, the algorithm finds the k closest neighbors in the feature space and assigns the class based on their majority vote. A larger k makes the model more generalized, while a smaller k may cause overfitting. KNN is simple and intuitive, but can be slow on large datasets and is sensitive to feature scaling.

accuracy : 0.8482142857142858



Conclusion

Conclusion

The project focused on predicting whether the first stage of a Falcon 9 launch would successfully land, using multiple machine learning models trained on historical launch data, with the Decision Tree classifier achieving the best performance.

- Launch success rates have consistently improved over time, particularly after 2017, indicating increasing reliability as SpaceX gained more experience.
- Lower payload masses (below 4000 kg), specific launch sites (such as KSC LC-39A / CCAFS LC-40), and certain configurations (including the FT booster and SSO orbits) were strongly associated with higher landing success rates.
- Geographic analysis showed that launch sites are strategically located close to coastlines and transportation infrastructure, while remaining distant from urban areas, supporting safe and efficient launch operations.

Thank you!